Materials, constructs and primers

A procurement and sequence reference for the Cell_x_Cell CLASP-seq workflow.

Working document. Red **TO COMPLETE** fields identify missing or conflicting author inputs. Complete the fields affecting a procedure before using it at the bench. Recorded settings are not a claim of experimental optimization.

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01 Using this reference

Use this inventory with the main protocol and preparation guides. Recorded identities are retained from the laboratory notes; red fields identify information needed to obtain, prepare or verify a material. Internal material names are not repository accessions or portable catalog numbers. [D1 sections 3, 6.3-6.4]

TO COMPLETE - C14
Add the method authors and contact, released material-list version, distribution routes and applicable reuse or transfer terms.

Confirm each construct before ordering primers
The notes contain a documented HLA primer conflict. CZP31 is obsolete for the current HLA construct; CZP31.0 is the registry-directed replacement, subject to confirmation against the actual released construct. The sequences below are a reference record, not a completed synthesis order. [S1 p.2; S10 p.1]

02 Cells and genetic materials

Material recorded	Role	Release information
Jurkat: CD8+, FAS-, TCR-	TCR recipient cells. [S1 p.1]	TO COMPLETE: C01 - exact derivative, source/accession, engineering history, authentication, complete medium and culture conditions.
K562	HLA-peptide recipient cells. [S1 p.1]	TO COMPLETE: C01 - exact derivative, source/accession, authentication, complete medium and culture conditions.
HEK293FT	Packaging cells. [S4 p.1]	TO COMPLETE: C01/C02 - source, passage limits, complete medium and packaging SOP.
MZ-1; TCRAFT-compatible vector	TCR transfer vector named in the overview. [S1 p.1]	TO COMPLETE: C01/C02 - annotated sequence, distribution route and relationship to the MZ-01 label in the optional TCRAFT notes.
CZ998E, CZ999E; receiver CZ4000E	HLA-peptide vector examples. [S3 p.1]	TO COMPLETE: C01 - full sequences, HLA allele, engineered substitutions, clone QC and distribution route.
VSVGwt, TMPS3, TMPS7	Packaging plasmids. [S4 p.1]	TO COMPLETE: C02 - exact identities, plasmid maps, availability and validated packaging recipe.

Stocks flagged for repair
The cloning notes identify HLA-B*51:01 and HLA-C*01:02 stocks as needing recloning: CZ4528/CZ4528R and CZ4532. Do not use these flagged stocks until corrected sequences and clone qualification are supplied. [S3 p.1]

What the construct manifest must contain

For each released construct, provide the full DNA and translated sequences, strand/orientation, signal peptides and linkers, engineered changes, associated biological identity, and primer-binding coordinates. Record the linked TCR α/β identity or the HLA allele and peptide explicitly. These records support both wet-lab verification and read assignment. [S1 p.1; S3 p.1]

TO COMPLETE - C01
Attach annotated construct files, a sequence-verified library manifest, corrected clone records and a concrete cell/plasmid access route. State any material-transfer or shipping requirements.

03 Reagents and consumables

Recorded material	Use	Information to complete
Lipofectamine; Viral boost	Cell-library generation. [S4 p.1]	TO COMPLETE: C02 - exact products, doses, stock formulations and complete transfection recipe.
Puromycin; polybrene	Selection and transduction. [S4 pp.1-2; S5 pp.1-2]	TO COMPLETE: C02 - products, stocks and confirmed selection schedule.
Miltenyi live/dead magnetic selection	Post-selection recovery. [S4 pp.1-2]	TO COMPLETE: C02 - exact kit and stage-specific procedure; expression antibodies/clones and controls where used.
Violet cell tracer	Label TCR cells. [S7 p.1]	TO COMPLETE: C03 - product, stock molarity, solvent, reconstitution and working dilution.
BS3; glycine	Stabilization and quench. [S8 p.1]	TO COMPLETE: C05 - product/form, fresh stock preparation, buffer pH and absolute glycine concentration.
PBS, HBSS, FBS, media, DMSO	Preparation, pairing and storage.	TO COMPLETE: C01/C04/C05 - formulations, supplements, relevant ion content, concentrations and storage.
KOD PCR system	Library preparation and targeted recovery. [S3; S9]	TO COMPLETE: C02/C09 - exact enzyme/product and stock concentrations for every component.
MinElute; gel extraction	DNA purification. [S1; S3]	TO COMPLETE: C02/C10 - exact products and stage-specific procedures.
SPRIselect/AMPure	DNA cleanup. [S1; S3]	TO COMPLETE: C09/C10 - exact bead product, procedure and stage-specific ratios; do not assume products are interchangeable.
Qubit; TapeStation	Concentration and fragment QC. [S1; S3]	TO COMPLETE: C10 - exact assays, assay ranges and dilution records.

Record manufacturer, catalog, lot and preparation date for each reagent used. A usable recipe specifies stock concentration, added volume, final volume, final concentration, buffer/solvent and storage interval. Labels such as '10× glycine' or 'dilute violet' do not supply these fields.

04 Capture, sequencing and equipment

Item	Recorded platform or requirement
10x capture	Chromium GEM-X Single Cell 3′ v4; source names CG000731 Rev B. This assay uses a custom capture and targeted-recovery workflow. [S1 pp.1-2]
10x release details	TO COMPLETE: C07 - exact kit/chip and lots, compatible instrument/firmware, custom-protocol supplement and annotated construct/read architecture.
Nanopore sequencing	Ligation sequencing amplicons V14, SQK-LSK114. The source names MinION sequencing and POD5 retention. [S1 p.3]
Nanopore release details	TO COMPLETE: C11 - pinned manual revision, instrument/flow-cell combination, auxiliary reagents, QC, acquisition and basecalling versions/settings.
Cell handling	Tissue-culture facilities, cell counter, temperature-controlled plate centrifuge and a defined shaker/rotator.
Sorting and capture	Sorter configured to retain intended cell pairs; compatible Chromium instrument; RT thermocycler.
Target recovery and QC	Thermocyclers, magnetic racks, DNA quantification and fragment-analysis equipment; selected Nanopore instrument.

TO COMPLETE - C06
Attach instrument models and portable settings, including sorter pressure/rate/mode, gate/control exports, collection setup and centrifuge rotor/brake. Define vessel and shaker geometry for pairing under C04.

Use institutional procedures for human-derived cells, lentiviral vectors, sorting and chemical handling. Include vector-system documentation and reagent SDS links with the released materials. Arrange the sorter, Chromium instrument and RT thermocycler for the same assay day; the notes do not establish an overnight pause after sorting. [S1 pp.1-3]

05 Core primer reference

Sequences and capitalization below preserve the supplied registry record. Confirm synthesis orientation, modifications, purification, stocks and construct-binding sites before use. This is the recorded core subset; it does not replace the completed oligo manifest. [S10 p.1]

Primer and recorded role	Sequence as recorded
CZP22 - peptide oligo-pool forward	CCGTGGTTACAGGAGGG
CZ5 - peptide oligo-pool reverse	GCTACCTCCACCGCCGGA
CZP47 - PCR1 forward	AAGCAGTGGTATCAACGCAGAG
CZP48 - PCR1 reverse	GCAGCGTCAGATGTGTATAAGAGACAG
CZP32UNI - overview TCR PCR2 primer	tcaacgcagagtaCGCGG
CZP31.0 - registry-directed HLA replacement; confirm construct	ATGGCGACGGGTTCAAGAACT
CZP31 - obsolete for current HLA construct; retained to identify conflict	CAGCCACCATGGCGACG

TO COMPLETE - C08
Publish a construct-to-primer compatibility table, resolve CZP31/CZP31.0, and specify when CZP32UNI or the TCRAFT-specific CZP160 applies. Supply the complete verified sequence and use record for any additional primer offered, including CZP160.

06 Indexed reverse primers and branch map

Each recorded indexed reverse primer has the form GCTTCC + sample index + CZP48. These indexes identify amplicon samples; they are not 10x partition barcodes or UMIs. [S10 p.1]

Registry ID and sample index	Full sequence as recorded
CZP48.01 Index: CTGCGAGTAT	GCTTCCCTGCGAGTATGCAGCGTCAGATGTGTATAAGAGACAG
CZP48.02 Index: ATAACTCCGG	GCTTCCATAACTCCGGGCAGCGTCAGATGTGTATAAGAGACAG
CZP48.03 Index: GCCCATGATT	GCTTCCGCCCATGATTGCAGCGTCAGATGTGTATAAGAGACAG
CZP48.04 Index: TATAGTGCGG	GCTTCCTATAGTGCGGGCAGCGTCAGATGTGTATAAGAGACAG

Assignments recorded in the overview

PCR2 branch	Overview primer labels	Required reconciliation
IN_TCR	CZP32UNI + CZP48.1	TO COMPLETE: C08 - confirm construct compatibility and .1/.01 alias.
EXP_TCR	CZP32UNI + CZP48.2	TO COMPLETE: C08 - confirm construct compatibility and .2/.02 alias.
EXP_HLA	CZP31 + CZP48.3	TO COMPLETE: C08 - replace obsolete primer in the overview after construct verification; resolve .3/.03 alias.
IN_HLA	CZP31 + CZP48.4	TO COMPLETE: C08 - same HLA conflict; resolve .4/.04 alias.

These are source labels awaiting reconciliation, not a finalized ordering or sample sheet. EXP_TCR and EXP_HLA are separate PCR2 products from the same experimental capture reaction. IN_TCR and IN_HLA come from independent input reactions. [S1 p.2]

TO COMPLETE - C07
Attach the annotated read architecture and example reads establishing sample-index, 10x barcode and any UMI coordinates/orientation. Index sequences alone do not establish barcode retention or correction rules.

TO COMPLETE - C09
Complete the PCR1 and PCR2 mixes, stock concentrations, template allocations, full cycling programs, cleanup methods and stage-specific pause points. Primer identity alone is insufficient to execute targeted recovery.

Source notes

Sources are the method authors' laboratory records, as transcribed in the supplied consolidation. The internal source files are not included in this download.

D1. CLASP-seq: the Cell_x_Cell TCR-pMHC pairing protocol. Consolidated author-review draft, 11 September 2026, sections 3, 6.3-6.4 and 12. This document preserves that draft's laboratory-note citations; underlying source files were not reinterpreted for this reference.

S1. Cell_x_Cell TCR_HLA-pep Pairing Assay V1. Author-supplied laboratory notes, 4 pages; citations retained from the consolidated author-review draft.

S3. PEP library cloning. Author-supplied laboratory notes, 5 pages; construct identities, cloning and flagged stocks.

S4. Generating Cell Lines for NxN Pairing Assay. Author-supplied laboratory notes, 2 pages; cell generation and qualification.

S5. Spinfection for Cell_x_Cell. Author-supplied laboratory notes, 3 pages; transduction and selection.

S7. Violet cell tracer staining. Author-supplied laboratory notes, 1 page; recorded tracer use and incomplete stock specification.

S8. BS3 Crosslinking. Author-supplied laboratory notes, 1 page; crosslinking and quench materials.

S9. PCR1, PCR2 for Cell_x_Cell Assay - STARTING VIRUS, 1st ROUND PCR. Author-supplied laboratory worksheet, 1 page; incomplete PCR formulation.

S10. CZ PRIMERS AND GENEBLOCKS - CZP1-. Author-supplied primer registry, 2 pages; core sequences, indexes and deprecation notes on page 1.

Protocol and resources